

Reaching the Actionable Radius: Physics-Guided Calibrated Localization of Methane Sources

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Abstract—We present a simple recipe that brings calibrated machine-learning localization of high-rate methane sources to the 50 m scale used for facility association in EPA super-emitter notifications. On a methane-configured testbed whose closed-form physics makes the *exact* Bayesian posterior computable for every scenario, we contribute an audit that measures, in meters, how much of the available information a calibrated localizer delivers: a conformal-wrapped deep localizer with a verified 90% coverage guarantee attains a 94 m median search radius where the exact posterior supports 6–7 m. Controlled elimination locates the cause — not supervision, and not the density estimator (a strengthened flow-based neural posterior estimator reaches only 63–71 m), but a classical computation (per-cell matched filtering) that pooled set encoders did not recover under any training signal we tested. Supplying it directly — two closed-form matched-filter maps as input channels and a zero-initialized residual head, 0.2% extra parameters, trained with ordinary labels — brings three architecture families to 44–49 m at unchanged coverage and millisecond latency; an ablation shows that once the maps are present, even distilling the exact posterior adds nothing. Under realistic wind-measurement error, exact plug-in inversion retains its guarantee only by returning essentially the entire site, while our learned localizers remain calibrated at 82–95 m, with uncertainty-aware input maps recovering a further 12 m. Calibrated, physics-guided learning is both sharper and more robust than inversion built on the same physics.

Index Terms—methane monitoring, conformal prediction, simulation-based inference, sensor networks, physics-guided machine learning

I. INTRODUCTION

Methane delivers roughly $80\times$ the 20-year warming potency of CO_2 and decays within about a decade, so rapid repair of oil-and-gas *super-emitters* is among the highest-leverage near-term climate actions. Detection has advanced rapidly — UNEP’s satellite alert system issued over a thousand notifications in its first two years [1] — and attribution is a critical downstream bottleneck between detection and operator investigation. Regulation suggests a concrete scale: EPA super-emitter notifications associate alerts with facilities at a 50 m radius [2] (program implementation currently extended to 2027), with analogous source-level duties in the EU [3]. We adopt 50 m as an operationally motivated, facility-scale localization target — a region that triggers a focused investigation, not yet a repair.

Continuous monitoring with sparse fixed point-sensor networks is where such localization can happen: a handful of

cheap masts around a facility, always on, catching the plume as it sweeps across them under naturally meandering wind. Machine learning supplies millisecond inference, and split conformal prediction [4] supplies a guarantee: a region provably containing the source 90% of the time. **The gap:** coverage guarantees frequency, not usefulness. Two questions had no quantitative answer: (i) how close is a calibrated region to the smallest region the sensor data support under the modeling assumptions, and (ii) what mechanism limits it, and what closes it? Standard diagnostics for amortized inference test whether a posterior is calibrated [5]; they cannot measure its distance from the exact posterior at matched calibration.

Contributions.

- **C1 — An exact-posterior audit.** A methane testbed (CH₄-T) whose closed-form physics makes the true Bayesian posterior of every scenario computable, so conformal efficiency can be measured in meters against a model-based information reference (6.2 m at matched empirical coverage) and the 50 m facility scale.
- **C2 — A diagnosis by elimination.** Controlled tests of five rival explanations — including a strengthened flow-based neural posterior estimator and, via a reusable teacher-conformalization diagnostic, the supervision itself — identify the dominant bottleneck in our experiments: a classical computation (per-cell matched filtering) that pooled encoders did not recover under any tested training signal.
- **C3 — A minimal recipe that reaches the scale.** Two closed-form matched-filter maps plus a zero-initialized residual head (0.2% extra parameters, ordinary label training) bring three architecture families to 44–49 m median radius at unchanged coverage and latency, and remain calibrated under wind-measurement error that reduces exact plug-in inversion to site-sized regions.

The components are deliberately classical; the ablation record (Appendix) shows the composition is minimal — notably, distilling the exact posterior, helpful without the maps, adds nothing once they are present. The exact posterior’s irreplaceable roles are measurement and diagnosis, not supervision.

II. RELATED WORK

Conformal prediction provides distribution-free coverage [4]; efficiency is the open half of the problem, and we

Figure 1 placeholder: wind-swept plume, masts, and true source on the 500 m site, with median 90% regions drawn to scale — exact-posterior reference (~ 7 m), our recipe (44–49 m), labels-only baseline (94 m), and the 50 m facility-association radius.

Fig. 1. The opportunity, to scale (placeholder).

contribute an audit that scores it against the exact Bayesian reference at matched, verified coverage. **Simulation-based inference** amortizes posteriors with neural density estimators [6], [7] at simulation budgets matching ours [8]; its diagnostics test calibration [5], and our audit adds the efficiency question while placing a representative flow-based estimator inside the comparison. **Source-term estimation** inverts dispersion models via optimization or MCMC at minutes-to-hours per event [10]; deep localizers give millisecond inference but lacked instruments certifying how much information they extract. **Physics-guided ML** injects domain structure through features, architectures, or teachers [9]; our input maps are the forward model’s sufficient statistics under CH4-T assumptions — classical matched-filter detection theory supplied as spatial channels. **Controlled-release evaluation** (METEC) scores continuous-monitoring systems on detection, quantification, and equipment-scale localization [11]; CH4-T mirrors that task structure at super-emitter rates with exact ground truth, complementing — not replacing — field validation.

III. METHODOLOGY

A. Testbed and exact-posterior audit (C1)

CH4-T places a source of unknown location and rate ($10\text{--}500\text{ kg h}^{-1}$, log-uniform, bracketing the 100 kg h^{-1} super-emitter threshold) on a 500×500 m site sensed by 4–12 masts (ppm readings, 0.2–3 ppm noise, dropout). The plume is a steady Gaussian under known Pasquill–Gifford stability with Briggs open-country dispersion (coefficients verified against published tables; mass flux conserved to 4×10^{-6}), driven quasi-statically by a known fluctuating wind record ($1\text{--}8\text{ m s}^{-1}$, AR(1) meander; steps read as 1-min averages, a 30-min window). Because the response is closed-form and linear in rate, the exact posterior over a 64×64 grid is computable per scenario (adaptive rate quadrature, float64, normalization error $\sim 10^{-12}$). A blocking gate requires conformal prediction applied to the exact posterior itself to self-calibrate — a gate that caught two implementation bugs, including a tie-handling error at degenerate scores whose fix we report for reuse. All models share identical data (10k/1k/2k/2k splits, the modal SBI budget [8]), D4 symmetry augmentation, and an identical randomized-HPD split-conformal wrapper at verified 90% coverage. Regions are reported as equal-area circle radii; the

TABLE I
MEDIAN 90% CONFORMAL REGION RADIUS (M); RANGE OVER 3 SEEDS; COVERAGE 0.89–0.92 IN EVERY CELL. REFERENCES: EXACT-POSTERIOR 6.2–7.0 M, FACILITY-ASSOCIATION SCALE 50 M.

Encoder	Labels only	+ Physics maps
DeepSets	94–98	47.1–47.7
GNN	55–58	[43–45][†]
Set Transformer	69–75	[45–49][†]

[†]Label-target replications in progress; shown values are from posterior-distillation targets, which the ablation (Appendix) finds equivalent to labels on DeepSets.

exact-posterior reference is 7.0 m at nominal 90% (empirical 0.95) and 6.2 m at matched empirical 0.89.

B. Diagnosis by elimination (C2)

A DeepSets localizer trained on one-hot labels is calibrated (0.89) at a 94 m median radius. Eliminated explanations, each by a controlled experiment (full record in the Appendix): *supervision content* — distilling the exact posterior (tempered; raw targets fail) improves to 74 m, yet auditing the tempered target itself shows it supports 11.7 m, so approximation, not supervision, binds; *target sharpness* — annealing toward the raw posterior worsens (87 m); *loss geometry* — a Sinkhorn transport loss fails (243 m, implementation-scoped); *the density estimator* — a representative flow-based neural posterior estimator, strengthened after adversarial review of our own implementation ($2.5\times$ budget, widened head, full-band conditioning), reaches 62.8–71.2 m, and we verify its spline head can express 4.4 m regions, so its blur is not head-structural. What survives: localization requires correlating each candidate cell’s space–time signature against all readings — per-cell matched filtering — which pooled set representations did not recover under any tested signal or budget.

C. The recipe (C3)

For any base localizer (Fig. 2): **(1)** compute per cell c the sufficient statistics $a_c = \sum g_c y$, $b_c = \sum g_c^2$ of the linear-Gaussian model (g_c the unit-rate response) and feed two 64×64 channels: the matched-filter statistic $z_c = a_c / (\sigma \sqrt{b_c})$ and $\log b_c$ — milliseconds, closed form, computed from the network’s own inputs (no new information, only a computation); **(2)** add a 3-layer zero-initialized convolutional head (~ 10 k parameters, 0.2% of the base) summed with the localizer’s logits, so at step 0 the model is the unmodified baseline and every comparison is controlled by construction; **(3)** train with ordinary smoothed-label cross-entropy; **(4)** wrap with unchanged split conformal. Neither ingredient suffices alone: the conformalized classical statistic without the network returns essentially the entire site (it is overconfident-wrong, and calibration must be learned), and the network without the maps delivers 94 m. Physics computes the evidence; the network weighs it honestly; conformal certifies the result.

Figure 2 placeholder (full width): method pipeline in two bands. Inference band (milliseconds): observed readings + wind $\rightarrow$ matched-filter maps $z, \log b \rightarrow$ any localizer + zero-init conv head $\rightarrow$ split-conformal 90% region. Side note: training uses ordinary labels; posterior distillation (dashed, optional) is redundant given the maps (Appendix).

Fig. 2. Physics-guided calibrated localization (placeholder).

TABLE II

A REPRESENTATIVE FLOW-BASED NPE HEAD ON THE SAME ENCODER, DATA, AND AUDIT; 3 SEEDS PER ROW.

Model (DeepSets encoder)	Radius (m)	Coverage
Flow-NPE (strengthened)	62.8–71.2	0.90–0.91
Flow-NPE + maps (+ distill.)	55.5–56.4	0.89–0.90
Ours (grid head + maps)	47.1–47.7	0.89

IV. RESULTS

A. One recipe, three architectures

The maps improve every architecture (Table I); the spread across encoders collapses (94-vs-56 without maps, ~ 49 -vs-44 with them), so the gains are available to whatever architecture a practitioner already has. Against the distillation-trained baseline the median radius reduction is 38% (bootstrap 95% CI [37.1, 39.3]; 91% of scenarios improved). Point accuracy is essentially unchanged: the maps improve the *shape of uncertainty*, which region size measures. Operationally: 51% of individual test cases fall below 50 m (median 49 m); the 75th/90th percentiles are 112/170 m; performance is mildly worse at the highest rates (55 m median at 250–500 kg h^{−1}) — the median case reaches the facility scale, and the upper tail defines the remaining challenge.

B. Comparison with a modern flow-based estimator

The flow head is a genuine lever — it beats posterior distillation with no physics at all — and the maps transfer onto it (Table II). The convolutional head reading the maps *spatially* extracts more than the flow’s parametric conditioning at identical information. Our approach is compared against, and complements, a modern estimator rather than being superseded by it.

C. Robust where plug-in inversion is not

Real anemometers err. Under a 10°/10% observation error, plug-in inversion at the measured wind assigns numerically negligible (below float64) posterior mass to the true source in 30% of scenarios; split conformal preserves its guarantee exactly as theory requires — by returning essentially the entire site (Table III). A global noise-inflation robust variant, tuned on calibration data, does not help: per-step wind error is not absorbable by a scalar noise scale. The learned localizers, trained through the same error, remain calibrated at useful sizes, and

TABLE III

WIND MEASURED WITH REALISTIC ERROR (10° DIRECTION, 10% SPEED); TRAINING, CALIBRATION AND TEST ALL UNDER THE SAME ERROR DISTRIBUTION; 3 SEEDS FOR LEARNED ROWS.

Pipeline	Coverage	Radius
Plug-in inversion (measured wind)	1.00	282 m (full site)
Plug-in, inflated-noise robust variant	1.00	282 m
Ours, deterministic maps	0.91	93–95 m
Ours, ensemble maps	0.88–0.90	82–85 m

ensemble maps — the mean and spread of the matched-filter maps over eight draws from the wind-error model — recover a further ~ 12 m (sharper on 74% of scenarios). Under the modeled wind-measurement error, learned physics-guided localization is more robust than plug-in inversion built on the same physics; a wind-marginalized exact reference is future work.

V. LIMITATIONS

CH4-T is an instrument, not a deployment: single steady sources (no zero-leak or multi-source scenarios), flat terrain without equipment wakes, known stability class, quasi-static stepping, in-distribution marginal coverage, and a 64×64 grid whose ~ 7.8 m cells sit near the oracle reference scale (a grid-refinement study is pending). The 6–7 m reference is a property of CH4-T’s generative assumptions, not an atmospheric limit. Half of individual cases still exceed 50 m, and the upper tail (p90 = 170 m) is unresolved. The architectural claim is scoped to tested encoders and budgets; the transport-loss null is implementation-scoped. Field claims await controlled-release validation — comparisons here are exact regardless, because all models share one verified simulator.

VI. CONCLUSIONS

On a testbed with exact Bayesian ground truth, a calibrated deep localizer delivered a tenth of the sharpness its sensor data support; controlled elimination traced the gap to a single classical computation, and supplying that computation — two matched-filter maps and a 0.2%-parameter head, ordinary labels, nothing else — brought three architectures to the 50 m facility-association scale at unchanged cost, ahead of a strengthened flow-based estimator, and calibrated under wind error that reduces exact inversion to site-sized regions. The constructive message: simulators make exact posteriors

available as *instruments* (audit and diagnosis) even when, as our ablation shows, they are unnecessary as supervision; and classical signal processing still knows computations that networks need not rediscover. We hope the evidence standard — exact-posterior audits, blocking self-calibration gates, one-change-at-a-time controls with documented nulls — travels furthest.

APPENDIX

APPENDIX: ABLATION AND CONTROL RECORD

All rows: DeepSets encoder unless noted, identical data and conformal harness, median 90% region radius (m), 3 seeds where marked.

TABLE IV
WHAT MATTERS AND WHAT DOES NOT (CLEAN-WIND TRACK).

Configuration	Radius (m)	Coverage
Labels only (baseline)	94–98 ^{3s}	0.89
Posterior distillation, raw teacher	134	0.90
Posterior distillation, tempered teacher annealed toward raw	73–74 ^{3s}	0.90
Sinkhorn transport loss	87	0.89
	243	0.91
Maps + head, labels	47.1–47.7^{3s}	0.89
Maps + head, distillation targets	48.1–49.3 ^{3s}	0.89
z channel only (labels+distill.)	55.1	0.89
$\log b$ channel only	73.0	0.92
Classical statistic only, conformalized	282 (full site)	1.00
Distillation-target reference (audited)	11.7	0.90
Exact-posterior reference	6.2–7.0	0.89–0.95

Reading. Distillation is worth ~ 20 m *without* the maps (94 $\rightarrow$ 74) and nothing *with* them (47.4 vs 48.7 — labels slightly better); neither sharper targets nor a transport loss substitute for the maps; both channels contribute, the matched filter dominating; and the classical statistic alone is overconfident-wrong, forcing full-site regions — the learned calibration is load-bearing. Together: the binding constraint was the computation itself; once supplied, supervision choice is secondary. Rate-stratified performance (median radius / fraction below 50 m): 10–30 kg h^{−1}: 44.8 m / 54%; 30–100: 44.5 m / 55%; 100–250: 50.3 m / 49%; 250–500: 55.2 m / 41%.

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